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Processando config 28, seed 0, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

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smultin_MultinomialBitsOver test:

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N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = 1 / 32768
EColl = n^2 / (2k) = 128
Hashing = TRUE

Collision test
CollisionOver: density = n / k = 1 / 32768.00
Expected number of collisions = Mu = 128.00

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Results of CollisionOver test:

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POISSON approximation :
Expected number of collisions = N*Mu : 256.00
Observed number of collisions : 273
p-value of test : 0.15

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Total number of cells containing j balls

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j = 0 : 549739036945
j = 1 : 16776670
j = 2 : 273
j = 3 : 0
j = 4 : 0
j = 5 : 0

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CPU time used : 00:00:04.48

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Generator state:
16777216 bits have been read.

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.08

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.93
p-value of test : 0.18

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 10.25
p-value of test : 0.17

Normal statistic for number of jumps : -50.42
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.05

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.95
p-value of test : 0.17

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.91
p-value of test : 0.82

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.044
p-value of test : 0.14

Kolmogorov-Smirnov- statistic = D- : 2.71e-3
p-value of test : 0.9907

Anderson-Darling statistic = A2 : 1.56
p-value of test : 0.16

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 4.12
p-value of test : 0.25

Global longest run of 1 : 27.00
p-value of test : 0.06

CPU time used : 00:00:00.05

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 84.74
p-value of test : 0.74

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 25.41
p-value of test : 0.28

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -2.18
p-value of test : 0.99

CPU time used : 00:00:00.01

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.21
p-value of test : 0.58

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 1.24
p-value of test : 0.11

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 177.91
p-value of test : 0.47

CPU time used : 00:00:00.01

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 302.71
p-value of test : 0.53

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 473.26
p-value of test : 0.69

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1


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-----  
Normal statistic          :   -0.86  
p-value of test          :    0.80  
-----
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CPU time used             :   00:00:00.01
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Generator state:  
269499904 bits have been read.
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin
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sstring_AutoCor test:
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-----  
N = 1, n = 16777154, r = 0, s = 32, d = 2  
-----
```

```
-----  
Normal statistic          :    0.95  
p-value of test          :    0.17  
-----
```

```
CPU time used             :   00:00:00.00
```

```
Generator state:  
286277088 bits have been read.
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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin
```

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sstring_Run test:
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-----  
N = 1, n = 3355443, r = 0, s = 32  
-----
```

```
-----  
Total number of 1 runs: 3355445
```

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Number of degrees of freedom : 36  
Chi2 statistic for number of runs : 30.77  
p-value of test : 0.72  
-----
```

```
Total number of bits: 13414080
```

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Normal statistic for number of bits : -1.48  
p-value of test : 0.93
```

CPU time used : 00:00:00.04

Generator state:
299691168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 1.58
p-value of test : 0.66

CPU time used : 00:00:00.07

Generator state:
316468384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333159584 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	39.20
p-value of test	:	0.59

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	43.75
p-value of test	:	0.40

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	54.16
p-value of test	:	0.80

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	36.49
p-value of test	:	0.45

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	22.04
p-value of test	:	0.46

CPU time used : 00:00:00.05

Generator state:
349936800 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	79.22
p-value of test	:	0.86

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	91.41
p-value of test	:	0.56

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	484.03
p-value of test	:	0.81

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	73.61
p-value of test	:	0.90

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	40.68
p-value of test	:	0.82

CPU time used : 00:00:00.03

Generator state:

366714016 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	146.37
p-value of test	:	0.15

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	121.57
p-value of test	:	0.71

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	161.17
p-value of test	:	0.48

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	136.40
p-value of test	:	0.33

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	97.01
p-value of test	:	0.26

CPU time used : 00:00:00.02

Generator state:

383490816 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:05.94
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
5 LinearComp	1 - eps1
22 MatrixRank, 320 x 320	eps
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All other tests were passed